PALAWAN STATE UNIVERSITY
TAYTAY CAMPUS

GINTONG ARAW: OSCA – SENIOR CITIZEN SERVICES

A Flutter-Based Mobile Application

A Project Documentation

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1. INTRODUCTION

Senior citizens are an important part of every community and deserve convenient access to information, benefits, and services provided by their local government offices. The Office of Senior Citizens Affairs (OSCA) plays an important role in assisting senior citizens by providing information about available benefits, discounts, identification, and other related services. However, accessing these services and keeping track of important information may require individuals to visit the office or communicate directly with authorized personnel.

Gintong Araw: OSCA – Senior Citizen Services is a proposed mobile application designed to provide senior citizens with convenient access to information and account-related services through a smartphone. The application provides a centralized interface where users can view their senior citizen identification information, available benefits and discounts, transaction records, and important account details.

The project will use the Flutter framework for mobile application development. Flutter is an open-source user interface toolkit created by Google that uses the Dart programming language (Flutter, n.d.-a; Dart, n.d.). Flutter and Dart, together with Visual Studio Code as the development environment and Figma as the design tool, form the technology stack chosen for Gintong Araw.

Flutter is the appropriate choice for this project for three main reasons. First, it offers cross-platform capability: a single codebase can produce applications for both Android and iOS, allowing the application to reach more senior citizens without building two separate apps. Second, Flutter provides a rich set of ready-made

widgets, including Material and Cupertino components, which are used to build forms, cards, buttons, lists, and navigation bars (Flutter, n.d.-c). Third, Flutter is supported by a robust developer community, with extensive official documentation and a large collection of open-source packages available through pub.dev (pub.dev, n.d.), which makes it easier to find solutions and learning resources during development.

The application includes account access features such as login, account registration, password recovery, verification code, and password reset. After accessing their account, users can navigate through different sections, including Home, Benefits, Verification, Transactions, and Profile.

Gintong Araw also provides information about medicines and health discounts, frequently asked questions, contact information for OSCA, notifications, language preferences, accessibility options, and privacy settings. These features are designed to make important senior citizen information easier to access and understand.

The interface designs prepared in Figma serve as the visual reference for the application's development. This documentation describes the functions represented by the available screen designs and distinguishes interface features from services that still require actual implementation.

2. PROJECT OVERVIEW

Gintong Araw aims to improve access to senior citizen information and OSCA-related services through a mobile application. It provides an organized interface where registered users can view their digital senior citizen identification, review available benefits, monitor transaction information, and manage their personal account details.

The application is organized into several sections to make navigation easier for senior citizens. The main navigation consists of Home, Benefits, Verification, Transactions, and Profile. Each section provides access to specific information and related functions. The application also includes an OSCA Staff interface intended to support authorized personnel in reviewing senior citizen records, checking verification requests, and monitoring recent activities.

Flutter will enable cross-platform development for Gintong Araw. Because Flutter builds Android and iOS applications from one Dart codebase, the same screens, navigation, and application logic can run on both platforms. This reduces development time and cost, since the screens do not have to be written and maintained twice. Initial development and testing will focus on Android, while the same codebase can later be built for iOS with limited additional work.

The project plans to utilize the following key features of Flutter:

- **Hot reload:** allows code changes to appear in the running application almost instantly without a full restart, which speeds up layout adjustments and the correction of interface issues (Flutter, n.d.-b).

- **Pre-designed widgets:** Material widgets such as Scaffold, AppBar, Card, ListTile, TextFormField, and NavigationBar are used to build the screens in a consistent way.
- **Responsive user interfaces:** layout widgets such as Row, Column, Expanded, and Flexible, together with MediaQuery, allow the screens to adapt to different phone sizes and support adjustable text size.
- **Reusable components:** custom widgets, such as benefit cards and transaction items, can be written once and used across several screens.

2.1 Technology Stack

Technology	Purpose in the Project
Flutter	Framework used for developing the mobile application's user interface and screens for Android and, in the future, iOS.
Dart	Programming language used for Flutter application development.
Visual Studio Code	Development environment used to manage and edit the application source code.
Figma	Design tool used to prepare the application's screen layouts and visual references.
Android Device / Emulator	Target environment for initial application testing and interface evaluation.

2.2 Why Flutter Is Appropriate

Flutter is appropriate for the Gintong Araw project because it provides a widget-based approach to mobile application development. Its reusable widgets can be used to create common interface elements such as buttons, text fields, cards, navigation bars, icons, and forms (Flutter, n.d.-c).

The application contains multiple connected screens, including account registration, login, verification, benefits, transactions, and profile management. Flutter allows these screens to be organized using reusable components, helping maintain consistency throughout the application.

Flutter also supports Android and iOS development through a shared codebase. This provides an opportunity for future platform expansion while allowing the initial development and testing to focus on Android devices. In addition, the hot reload functionality helps developers review interface modifications and identify layout issues during development, and the active Flutter community provides documentation, tutorials, and packages that support students and new developers.

2.3 Project Scope and Limitations

The project focuses on the development of the Gintong Araw mobile application interface and its planned user navigation. The current application screen designs include account registration, login, password recovery, verification, digital senior citizen identification, benefits information, transaction records, profile management, and application settings. The application also includes an OSCA Staff interface showing registered senior citizen statistics, verification requests, and recent activities.

The current screen designs demonstrate the intended appearance and workflow of these features. However, the availability of a connected database, backend service, live authentication, OTP delivery, and real-time synchronization must be verified through the actual application implementation.

Without an operational backend, information may be limited to sample or locally maintained data. Live account verification, centralized senior citizen records, transaction synchronization, and actual communication with OSCA personnel would require additional development and supporting services. The project documentation therefore distinguishes between features represented in the interface and features that have been fully implemented and tested.

3. TARGET AUDIENCE

The primary target audience of Gintong Araw consists of senior citizens who need convenient access to information about their identification, benefits, discounts, and related services. Specifically, the application is intended for Filipino senior citizens aged 60 years and above, the age recognized for senior citizens under the Expanded Senior Citizens Act of 2010, who are registered with or seeking registration with their local OSCA.

Many of these users may have limited or varying experience with smartphones. For this reason, the application is designed to provide an understandable mobile interface with readable text, simple labels, adjustable text size, and both English and Filipino language options.

Flutter's versatility and cross-platform compatibility allow the application to cater to a broader user base. Because one codebase can serve both Android and iOS, senior citizens and their families are not limited by the brand of smartphone they use once iOS support is added. The initial development and testing will focus on Android users, and future platform support may be considered depending on the project's requirements.

The intended users include:

- **Senior Citizens:** Individuals who can use the application to view their digital senior citizen identification, check available benefits and discounts, review transaction information, and manage their account details.
- **New Users:** Senior citizens who need to create an account and provide the required information to access the application's services.

- **Registered Users:** Existing account holders who can sign in, access their dashboard, view benefits, and manage their personal information.
- **OSCA Staff:** Authorized personnel who may use the administrative interface to review senior citizen records, verify identification information, and monitor recent application activities.

4. FEATURES

Gintong Araw provides different application features designed to make senior citizen information and account services more accessible. The mobile interface uses a consistent design with organized forms, information cards, buttons, and bottom navigation. These interface elements help users move between the application's different sections.

4.1 Flutter's Widget-Based Architecture

In Flutter, everything on the screen is a widget, and complex screens are built by combining smaller widgets (Flutter, n.d.-c). This architecture will be used to create the application's interface and functionality efficiently. Each screen of Gintong Araw will be built around a Scaffold widget, with an AppBar for the screen title and a NavigationBar for moving between Home, Benefits, Verification, Transactions, and Profile.

Forms such as Login, Create Account, and Change Password will use Form and TextFormField widgets with validators to check incomplete or invalid entries. Lists of benefits, transactions, and notifications will use ListView, Card, and ListTile widgets. Screen changes will be handled by Flutter's Navigator. Static screens, such as About Gintong Araw, can use StatelessWidget, while screens that change based on

user input, such as forms and settings, can use StatefulWidget. Repeated elements, such as benefit cards and transaction items, will be created as reusable custom widgets to keep the design consistent and reduce repeated code.

4.2 Planned Flutter Packages and Tools

The application will initially use Flutter's standard Material widgets and Navigator or named routes. The following packages from pub.dev are planned to enhance the application's features when the related functions are implemented (pub.dev, n.d.).

Package / Tool	Planned Purpose
Material widgets and Navigator (built-in)	Interface components and navigation between application screens.
provider	Managing application state such as the logged-in user, language, and text size.
shared_preferences	Storing simple local settings such as language and accessibility preferences.
intl and flutter_localizations	English and Filipino language options and date and currency formatting.
qr_flutter	Generating a QR code on the Digital Senior ID.
mobile_scanner	Scanning the QR code during staff identification verification.
image_picker	Attaching a screenshot or receipt in Report a Problem.
flutter_local_notifications	Showing benefit reminders and account notifications.
flutter_tts	Providing spoken guidance as an accessibility option.
firebase_auth and cloud_firestore	Optional account authentication, OTP, and database services if a backend is implemented.

The final package selection will depend on the features actually implemented in the application. Packages for backend services will be added only when those services are implemented.

4.3 Senior Citizen Application Features

Feature	Description
Welcome Screen	Introduces Gintong Araw and provides access to the account login and registration options.
Login	Provides an interface where registered senior citizens can enter their mobile number or OSCA ID and password.
Create Account	Allows new users to enter their full name, OSCA ID, mobile number, and password to create an account.
Forgot Password	Provides an account recovery interface where users can request a verification code using their registered mobile number.
Verify Code	Allows users to enter the six-digit verification code sent to their registered mobile number.
Reset Password	Provides an interface for creating and confirming a new account password.
Home Dashboard	Displays a welcome message, senior citizen profile information, registration status, available benefits, quick access options, and recent transactions.
Benefits	Displays available senior citizen benefits and discounts, including medicines and health, dining, and transportation.
Medicines and Health	Provides information about medicine and health-related discounts, eligibility requirements, and instructions for using the benefit.
Verification	Displays the user's verification status and provides access to the digital senior citizen ID. It also includes an interface for authorized staff verification.
Digital Senior ID	Displays the senior citizen's identification details, including name, OSCA ID, date of birth, age, barangay, registration status, and verification information.
Verification Result	Displays the result of an identification verification process and the corresponding senior citizen information.

Feature	Description
Transactions	Displays the user's transaction history, total savings, transaction dates, establishments, and discounts received.
Transaction Details	Provides detailed information about a selected transaction, including purchase information, gross amount, senior discount, amount payable, and transaction reference.
Profile	Displays the user's personal information and account verification status.
Personal Information	Allows users to view and update their full name, date of birth, home address, and mobile number.
Navigation and Routes	Provides organized navigation between Home, Benefits, Verification, Transactions, and Profile.

4.4 Settings and Help Center Features

Feature	Description
Settings and Help Center	Provides access to account settings, application preferences, and support services.
Edit Personal Information	Allows users to access the personal information editing interface.
Change Password	Provides a form for entering the current password and creating a new password.
Update Contact Information	Provides an interface for updating the registered mobile number and email address.
Account Security	Provides access to account protection and security-related options.
Help and FAQs	Displays frequently asked questions about senior citizen services, identification, benefits, and account recovery.
Contact OSCA	Provides a form for sending an inquiry to the user's local OSCA office.
Report a Problem	Allows users to report concerns by selecting an issue category and providing transaction details or a description. It also provides an option to attach a screenshot or receipt.
About Gintong Araw	Provides information about the application's purpose, features, and version information.

Feature	Description
Notifications	Displays updates about benefits, account activities, transaction information, and other application notifications.
Notification Settings	Allows users to manage notification preferences for benefit reminders, transactions, account security, and application updates.
Language Selection	Provides language options, including English and Filipino.
Accessibility Options	Provides interface preferences such as text size, higher contrast, spoken guidance, and confirmation of important actions.
Privacy and Security	Provides privacy controls, notification privacy, screen lock options, and privacy policy access.

4.5 OSCA Staff Features

Feature	Description
OSCA Staff Welcome	Provides a separate entry interface for authorized OSCA staff.
Admin Dashboard	Displays senior citizen registration statistics, including total registered seniors, verified accounts, and pending verification requests.
Verify Senior ID	Provides access to the senior citizen identification verification process.
Register Senior	Provides an interface for registering a senior citizen record.
Verification Queue	Displays senior citizen verification requests requiring staff review.
Recent Activity	Displays recent registration and verification activities within the staff interface.
Application Monitoring	Provides an overview of selected application activities and verification information.

5. TESTING AND EVALUATION

Testing will be conducted as the Gintong Araw application screens and functions are implemented. The primary objective of testing is to determine whether the application's interface displays correctly, navigation works as intended, and forms respond appropriately to user input. Testing will also focus on the usability of the application, especially for senior citizens who need readable text, understandable instructions, and accessible navigation controls.

5.1 Testing Frameworks and Tools

The Flutter ecosystem includes a built-in testing framework that supports three levels of testing (Flutter, n.d.-d). The following will be used to ensure the quality and reliability of the application:

- **Unit tests:** use the Flutter Test framework (flutter_test) to check single functions, such as input validation for the login and registration forms.
- **Widget tests:** use WidgetTester to check that screens, buttons, forms, and navigation elements appear and respond correctly without needing a physical device.
- **Integration tests:** use the integration_test package to run the complete application on an Android emulator or device and check full workflows, such as logging in and opening a transaction.
- **Debugging tools:** Flutter DevTools and the flutter analyze command will be used to inspect layouts, find errors, and detect code problems.

Flutter's support for automated testing allows tests to be written once and run again after every change using the flutter test command. This helps detect

problems early and confirms that new changes do not break existing screens. Testing libraries such as mockito can also be used to simulate data or services that are not yet available, making testing and debugging easier during development.

5.2 Testing Approach

- **Widget Testing:** Flutter widget testing tools will be used to check whether important interface components, such as buttons, forms, and navigation elements, appear and respond as expected.
- **Navigation Testing:** The application's navigation controls will be tested to determine whether Home, Benefits, Verification, Transactions, and Profile lead to their correct screens.
- **User Interface Testing:** The application will be checked for layout consistency, readable text, proper spacing, and correct display of interface elements.
- **Input Validation Testing:** Registration, login, password recovery, and profile forms will be checked for incomplete or invalid entries.
- **Verification Testing:** The verification process will be checked using valid and invalid identification information if the function is implemented.
- **Transaction Testing:** Transaction records and transaction details will be checked to determine whether the displayed information corresponds to the selected transaction.
- **Accessibility Testing:** Text size, contrast, language selection, and other accessibility options will be reviewed to determine whether the interface is understandable and usable.

- **Usability Evaluation:** Selected users may be asked to try the application and provide feedback about its clarity, navigation, and ease of use.

5.3 Evaluation Criteria

Criteria	What Will Be Checked
Functionality	Whether implemented buttons, forms, and navigation controls perform their intended actions.
Usability	Whether senior citizens can understand the labels, instructions, and application workflow.
Compatibility	Whether the interface displays correctly on the selected Android device or emulator.
Reliability	Whether the application operates without repeated crashes or unexpected behavior during testing.
Accessibility	Whether text size, contrast, and available interface preferences support usability.
Accuracy of Information	Whether the displayed senior citizen information, benefits, and transaction details are correct.
Security	Whether implemented account access and personal information protection mechanisms operate as intended.

Testing results, discovered issues, and corrections should be recorded after the application becomes runnable. Features that depend on external services should only be considered successfully tested after their actual implementation.

6. PROJECT TIMELINE AND BUDGET

The use of Flutter is expected to help shorten the development timeline of Gintong Araw. Hot reload shortens the cycle of editing and checking screens, reusable widgets reduce repeated work, and a single codebase removes the need for a separate development phase for each platform. A shorter development period also helps reduce overall project costs.

6.1 Project Timeline

The following table presents the proposed 15-week development phases for Gintong Araw. The schedule may change depending on development progress, testing results, and available project resources.

Phase	Schedule	Activities
Project Planning and Requirements	Weeks 1–2	Identify project objectives, intended users, application requirements, and necessary features.
UI Design and Planning	Weeks 3–4	Prepare and finalize the mobile application screen designs using Figma.
Flutter Development	Weeks 5–8	Set up the Flutter project and develop the application screens using Dart.
Navigation and Functionality	Weeks 9–11	Connect the screens and implement the selected application functions.
Testing and Evaluation	Weeks 12–13	Test navigation, forms, screen layouts, and application usability.
Documentation and Presentation	Weeks 14–15	Complete the project documentation and prepare the application demonstration.

6.2 Estimated Project Budget

The following table presents the estimated budget for the project. Actual costs may vary based on the resources used during development.

Item	Purpose	Estimated Cost
Development Tools	Flutter, Dart, Visual Studio Code, and Figma (free versions).	₱0.00
Internet Connection	Internet access for development and research.	₱1,500.00
Android Testing Device	Existing personal device used for application installation and testing.	₱0.00
Printing and Documentation	Printing and preparation of project documents.	₱500.00
Other Expenses	Additional project-related expenses.	₱500.00
TOTAL	**Estimated project expenses.**	**₱2,500.00**

6.3 Budget Assumptions

- Flutter, Dart, Visual Studio Code, and the free version of Figma have no license cost, so they are not treated as paid software expenses.
- Flutter's single codebase avoids the cost of building and maintaining separate Android and iOS applications, which lowers development effort and cost.
- The Android testing device is an existing personal device, so no purchase cost is included. Internet and device expenses are included only when applicable.
- Additional costs for hosting, database services, OTP delivery, notifications, and an Apple Developer account for future iOS release are not included and should be added only if these services are implemented.

- The budget should be updated whenever the project requirements change.

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8. DEVELOPER PROFILE



Name: Jerry B. Navarro

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School: Palawan State University – Taytay Campus

Project: Gintong Araw: OSCA – Senior Citizen Services

Technical Skills:

- Flutter and Dart mobile application development
- Mobile user interface design using Figma
- Widget-based screen layout, forms, and navigation
- Source code management using Visual Studio Code
- Basic application testing and debugging using an Android emulator or device